Theory of Basic Linear Algebra Concepts

Introduction

Linear algebra is an important part of mathematics. It is used in data science, computer graphics, machine learning, engineering, and many other fields. In this section, we study vectors, matrices, eigenvalues, decompositions, and their practical use. The aim of this theory is to explain these topics in simple and easy language with formulas and examples.

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1. Vectors, Operations, Distances, Norms

A **vector** is a quantity that has both magnitude and direction. A vector is usually written as a list of numbers.

Example:

If **A = (2, 3)**, it means the vector has two values.

Vector operations:

- **Addition:** add the corresponding values
- **Subtraction:** subtract the corresponding values
- **Scalar multiplication:** multiply every value by the same number

If **A = (2, 3)** and **B = (1, 4)**:

A + B = (3, 7)

A - B = (1, -1)

2A = (4, 6)

Distance tells us how far two vectors are from each other.

Formula:

Distance between A and B:

`d(A,B) = sqrt((x2 - x1)^2 + (y2 - y1)^2)`

Example:

Let A = (1, 2) and B = (4, 6)

Distance = sqrt((4 - 1)^2 + (6 - 2)^2)

= sqrt(3^2 + 4^2)

= sqrt(9 + 16)

= sqrt(25) = 5

A **norm** means the length or size of a vector.

Formula:

`||A|| = sqrt(x^2 + y^2)`

Example:

If A = (3, 4), then

||A|| = sqrt(3^2 + 4^2) = sqrt(9 + 16) = 5

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2. Unit Vector, Dot Product, Cross Product

A **unit vector** is a vector with length 1. It shows only direction.

Formula:

`u = A / ||A||`

Example:

If A = (3, 4), then ||A|| = 5

So unit vector = (3/5, 4/5)

A **dot product** gives a single number. It helps us measure similarity between two vectors.

Formula:

`A . B = a1b1 + a2b2 + a3b3`

Example:

If A = (1, 2) and B = (3, 4), then

A . B = (1×3) + (2×4) = 3 + 8 = 11

A **cross product** gives a new vector. It is mainly used for 3D vectors.

Formula:

`A x B = (a2b3 - a3b2, a3b1 - a1b3, a1b2 - a2b1)`

Example:

If A = (1, 0, 0) and B = (0, 1, 0), then

A x B = (0, 0, 1)

Dot product vs Cross product:

- Dot product gives a **number**
- Cross product gives a **vector**
- Dot product is used for angle and similarity
- Cross product is used for direction in 3D space

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3. Angle Between Vectors, Projection

The **angle between two vectors** tells us how much they point in the same direction.

Formula:

`cos(theta) = (A . B) / (||A|| ||B||)`

Example:

Let A = (1, 0) and B = (0, 1)

A . B = 0

So cos(theta) = 0

Therefore theta = 90°

This means the vectors are perpendicular.

A **projection** shows how much one vector lies in the direction of another vector.

Formula:

`proj_B(A) = (A . B / ||B||^2) B`

Example:

Let A = (2, 2) and B = (1, 0)

A . B = 2

||B||^2 = 1

Projection of A on B = 2(1, 0) = (2, 0)

This means only the horizontal part of A is taken.

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4. Matrix Types and Operations

A **matrix** is a rectangular arrangement of numbers in rows and columns.

Example:

[1 2]

[3 4]

Common matrix types:

- **Row matrix:** only one row
- **Column matrix:** only one column
- **Square matrix:** same number of rows and columns
- **Identity matrix:** diagonal values are 1, others are 0
- **Zero matrix:** all values are 0
- **Diagonal matrix:** only diagonal values may be non-zero

Matrix operations:

- Addition
- Subtraction
- Multiplication
- Transpose

Formula:

For addition:

` A + B = [aij + bij]`

For multiplication:

`Cij = sum(aik ,bkj)`

Example of addition:

If

A = [1 2; 3 4]

B = [5 6; 7 8]

Then

A + B = [6 8; 10 12]

Example of multiplication:

If

A = [1 2; 3 4]

B = [2 0; 1 2]

Then

AB = [4 4; 10 8]

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5. Eigenvalue and Eigenvector

An **eigenvector** is a special vector that does not change its direction when a matrix is applied to it. Only its length changes.

The number that changes its size is called the **eigenvalue**.

Formula:

`A x = lambda x`

Here:

A = matrix

x = eigenvector

lambda = eigenvalue

Example:

Let

A = [2 0; 0 3]

If x = [1; 0], then

Ax = [2; 0] = 2x

So, x is an eigenvector and 2 is the eigenvalue.

Use of eigenvalues and eigenvectors:

- data analysis
- image compression
- machine learning
- facial recognition
- system stability analysis

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6. Factorization / Decomposition (SVD, PCA, LDA)

Factorization or decomposition means breaking a matrix or dataset into smaller meaningful parts. It helps us understand data better and reduce complexity.

6.1 Singular Value Decomposition (SVD)

SVD breaks a matrix into three matrices.

Formula:

`A = U Sigma V^T`

Meaning:

U = left singular vectors

Sigma = singular values

V^T = right singular vectors

Example:

If we have a data matrix, SVD helps separate the most important information from less important information. It is useful in image compression and recommendation systems.

6.2 Principal Component Analysis (PCA)

PCA is used to reduce the number of features in data while keeping the most important information.

Formula:

`Z = XW`

Meaning:

X = original data

W = principal components

Z = reduced data

Example:

Suppose a student dataset has height and weight. PCA can combine these into one main feature that explains most of the variation.

6.3 Linear Discriminant Analysis (LDA)

LDA is used to separate data into classes. It tries to find the best line or direction that makes classes easy to distinguish.

Formula:

`W = argmax (between-class variance / within-class variance)`

Example:

If we have marks of two groups of students, LDA helps find a direction where the two groups look more clearly separated.

Difference between SVD, PCA, and LDA:

- **SVD:** matrix decomposition method
- **PCA:** reduces dimensions based on variance
- **LDA:** reduces dimensions based on class separation